

Modernization of the Magneto-Optical Structure of the Nuclotron to Study the Possibility of Measuring the Electric Dipole Moment of Light Nuclei with Preservation of Functions of the Booster of Polarized Beams for the NICA Collider

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Abstract—The aim of the proposed work is to determine the relationship between the Nuclotron and the NICA collider, as well as to modify the magneto-optical structure of the Nuclotron to create a synchrotron with two functions in one accelerator: a booster of polarized proton and deuteron beams for injection into the collider and an independent synchrotron-storage ring for conducting precision experiments on the study of the EDM of light nuclei and the search for an axion.

Keywords: spin, polarized beam, electric dipole moment

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INTRODUCTION

In order to solve the problem stated, it is proposed to consider two ways for upgrading the magneto-optical structure of Nuclotron 2.1 and Nuclotron 2.2, which, together with the main functions of the collider booster, should have additional functions that allow one to control and maintain polarization, as well as to conduct experiments to search for the electric dipole moment (EDM) of light deuteron and proton nuclei. In this case, all parameters of the magneto-optical structure should meet the conditions for polarization maintenance and control.

The fundamental issue in all the versions under consideration is achieving the intended luminosity of polarized beams, which obviously requires that there is no transmission of critical energy after the beam cools down. The latter is currently not included in the project, it is the main problem of the relationship between Nuclotron and the NICA collider. The most optimal way to solve this problem is to bring the critical energy of the collider beyond the limits of energy change in the acceleration cycle.

According to the current design, polarized beams from Nuclotron are injected into the collider at an

energy not exceeding the energy at which electron cools down in the collider, i.e., about 2–3 GeV. The latter condition is due to the required unconditional cooling of polarized beams after they are accumulated in the collider for achieving maximum luminosity in it. However, with this approach, the subsequent transmission of an accumulated cooled polarized proton beam through the critical energy in the NICA collider at $\gamma_{tr} \approx 7$ can lead to an increase in the phase volume of the colliding proton beam with a deuteron beam, thereby reducing their combined luminosity. The experimental measurements of the beam parameters carried out using a U-70 accelerator and the mathematical estimates of the growth of the proton beam emittance when the transmission of the critical energy yield a factor of the order of two. Therefore, this version should have an additional requirement where the proton transmission through the critical energy after cooling may be excluded. According to the requirements for the value of the “slip coefficient” from the point of view of collective instabilities, an increase in the critical energy of the collider to the value $\gamma_{tr} \approx 15$ –20 by the previously planned “resonant dispersion modulation” method should be considered. This could require the relatively simple introduction of an

additional family of focusing quadrupoles with a focusing field gradient increased by a relative value of $\sim 10\text{--}20\%$.

This method was developed by one of the authors of this work, became implemented in the Main Ring JPARC accelerator [1, 2], and has proven itself as a simple, reliable method. The same method for raising the critical energy was incorporated into the PS-2 accelerator project at CERN [3, 4] and the HESR ring in the FAIR project at the GSI, Germany [5]. In this regard, it is recommended that the “resonant” method is applied in both versions (2.1 and 2.2) of the Nuclotron magnetic optics.

BASIC PRINCIPLE FOR MEASURING THE EDM OF A CHARGED PARTICLE IN AN ACCELERATOR

The basic principle of EDM measurement is revolves around creating conditions under which the EDM signal is constantly accumulated as the beam is moving along the ring. For this, the orientation of the spin and longitudinal momentum vectors should be constantly co-directed $\mathbf{S} \rightarrow \uparrow \uparrow \mathbf{p}$ with an accuracy that determines the maximum possible sensitivity of the EDM measurement. The condition for the co-direction of the longitudinal momentum and spin is called spin frozenness or quasi-frozenness, depending on the implementation accuracy and method. The basic principle for measuring the accumulated EDM signal in the ring is derived from the T-BMT equation [6, 7] for spin precession in storage rings. In accordance with the Ehrenfest theorem, the T-BMT equations describe the classical behavior of the spin $\vec{S}$ of a charged particle with account for the assumed EDM:

$$\begin{aligned} \frac{d\mathbf{S}}{dt} &= \mathbf{S} \times (\boldsymbol{\Omega}_{\text{MDM}} + \boldsymbol{\Omega}_{\text{EDM}}), \\ \boldsymbol{\Omega}_{\text{MDM}} &= \frac{e}{m_0\gamma} \left\{ (\gamma G + 1) \mathbf{B}_\perp + (1 + G) \mathbf{B}_\parallel - \left(\gamma G + \frac{\gamma}{\gamma + 1} \right) \frac{\boldsymbol{\beta} \times \mathbf{E}}{c} \right\}, \\ \boldsymbol{\Omega}_{\text{EDM}} &= \frac{e\eta}{2m_0} \left(\boldsymbol{\beta} \times \mathbf{B} + \frac{\mathbf{E}}{c} \right), \quad G = \frac{g - 2}{2}. \end{aligned} \quad (1)$$

Here γ is the Lorentz factor; $\boldsymbol{\beta}$ is the relative velocity; c is the speed of light; e and m denote the particle charge and mass; G is the anomalous magnetic moment; g is the gyromagnetic ratio; $\boldsymbol{\Omega}_{\text{MDM}}$ is the spin precession frequency due to magnetic dipole moment (MDM); $\boldsymbol{\Omega}_{\text{EDM}}$ is the spin precession frequency due to EDM; η is the dimensionless coefficient determined in system (1) by the ratio $d = \eta e \hbar / 4mc$; $\mathbf{B} = \{\mathbf{B}_\perp, \mathbf{B}_\parallel\}$ and $\mathbf{E} = \{\mathbf{E}_\perp, \mathbf{E}_\parallel\}$ denote the magnetic and electric fields. Because the elements with a longitudinal magnetic field are not used here, it is accepted below that

$\mathbf{B}_\parallel = 0$. The longitudinal electric field $\mathbf{E}_\parallel = 0$ is also ignored due to the smallness of its contribution.

FROZEN SPIN METHOD

The frozen spin method [8] is based on the fact that, in the presence of so-called “magic” energy, the particle spin in external fields acquires a rotation frequency equal to the rotation frequency of the particle momentum along its orbit $\boldsymbol{\Omega}_{\text{E,B}}^p = \frac{e\mathbf{E}}{m\gamma\beta c} + \frac{e\mathbf{B}}{m\gamma}$. We extract frequency $\boldsymbol{\Omega}_{\text{E,B}}^p$ from $\boldsymbol{\Omega}_{\text{MDM}}$ to obtain a spin precession frequency relative to the direction of momentum $\boldsymbol{\Omega}_{\text{MDM}}^p$:

$$\boldsymbol{\Omega}_{\text{MDM}}^p = \boldsymbol{\omega}_E^p + \boldsymbol{\omega}_B^p, \quad (2)$$

where $\boldsymbol{\omega}_E^p = \frac{e}{m_0} \left(G - \frac{1}{\gamma^2 - 1} \right) \frac{\mathbf{E} \times \boldsymbol{\beta}}{c}$ is the frequency of spin precession relative to the momentum in the electric field and $\boldsymbol{\omega}_B^p = \frac{e}{m_0} G \cdot \mathbf{B}_\perp$ is the frequency of spin precession relative to the momentum in the magnetic field. Obviously, the frozen spin condition implies that the frequency difference is equal to zero: $\boldsymbol{\Omega}_{\text{MDM}} - \boldsymbol{\Omega}_{\text{E,B}}^p = 0$, i.e., $\boldsymbol{\Omega}_{\text{MDM}}^p = 0$.

In particular, the frozen spin condition in a fully electrostatic ring $\mathbf{B} = 0$ assumes the equality to zero $\boldsymbol{\omega}_E^p = 0$, which is fulfilled in the presence of “magic” energy:

$$G - \frac{1}{\gamma_{\text{mag}}^2 - 1} = 0. \quad (3)$$

In the E+B ring with magnetic and electric elements, the frozen spin condition is fulfilled under the same condition $\boldsymbol{\Omega}_{\text{MDM}}^p = \boldsymbol{\omega}_E^p + \boldsymbol{\omega}_B^p = 0$ that is satisfied in the presence of balance between the radial electric field E_r and the leading vertical magnetic field B_v within the following ratio:

$$E_r = \frac{GBc\beta\gamma^2}{1 - G\beta^2\gamma^2} \approx GB_v c\beta\gamma^2. \quad (4)$$

The E + B ring is applied to the case of deuterons because the G -factor for them has a negative value $G = -0.14$ and condition (3) is not fulfilled.

Further discussion requires the introduction of the generally accepted definition of spin tune. Its value determines the number of spin oscillations per revolution in an accelerator. The spin tune in the electrostatic ring ν_s^E , defined as the normalized spin precession frequency to the orbital frequency of the particle $\nu_s^E = \omega_E^p / \Omega_E^p$, is determined by the expression

$$v_s^E = \left(G - \frac{1}{\gamma^2 - 1} \right) \gamma \beta^2. \quad (5)$$

The spin tune $v_s^B = \omega_B^p / \Omega_B^p$ in the magnetic field relative to the momentum is calculated similarly:

$$v_s^B = \gamma G. \quad (6)$$

Thus, in the case of both protons and deuterons, we have a general idea of how to construct a ring for measuring the EDM, but this is implemented using different types of deflectors. The longitudinal electric field $E_{\parallel} = 0$ is also ignored due to the smallness of its contribution.

TRANSITION TO A STRUCTURE WITH A QUASI-FROZEN SPIN

Thus, condition (4) defines a relationship between the magnetic and electrostatic fields, assuming that both fields exist within a single element. The latter fact significantly complicates the design of an E + B deflector, thereby demanding the creation of a special ring for EDM studies. This eliminates the use of existing accelerators for EDM studies, which significantly complicates the problem and makes it financially difficult to build an accelerator specifically for this task. A similar statement can be made about a purely electrostatic ring for protons.

Therefore, we developed a quasi-frozen spin method that requires no specially oriented structure for studying the EDM [9]. The idea of such a structure is to separate the bending magnets and electric deflectors according to their location. For this purpose, we consider a spin equation in the electrostatic deflector and the bending magnet separately. In the previous paragraph, we defined the spin tune in the electric

field $v_s^E = \left(G - \frac{1}{\gamma^2 - 1} \right) \gamma \beta^2$ and in the magnetic field $v_s^B = \gamma G$.

Upon determining the ratio between them $K = v_s^E / v_s^B$, the dependence of this coefficient on energy is obtained. Figure 1 shows the ratio of the spin normalized frequency in the electric and magnetic fields to the particle momentum as a function of energy. Clearly, there is an energy range where the oscillation frequency of the MDM spin in the electric field is severalfold higher than in the magnetic field.

In this regard, we are interested in a deuteron beam energy range of ~250 MeV also because the maximum efficiency of the polarimeter using a carbon target for deuterons lies precisely in a range of 240–260 MeV. Due to this fact, the idea of a quasi-frozen structure (QFS) can be based on either a magnetostatic arch or an electrostatic arch of inverse curvature.

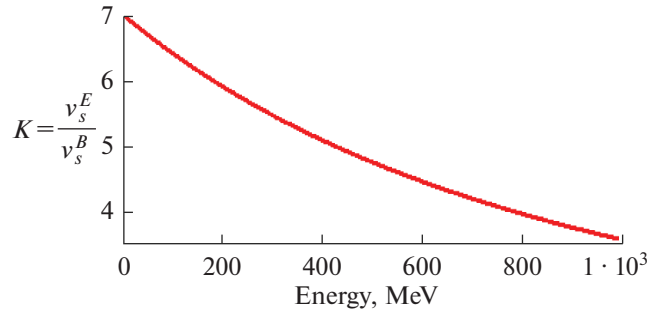


Fig. 1. Ratio v_s^E / v_s^B versus the beam energy.

ACTING NUCLOTRON STRUCTURE

Figure 2 illustrates the NICA accelerator complex consisting of Booster, Nuclotron, and Collider. The former two functionally act as an injector for heavy ions and polarized protons and deuterons into the collider. Initially, the magneto-optical structure of Nuclotron was shaped as close as possible to a circle with eight superperiods. Figure 3 shows the TWISS functions of one Nuclotron superperiod, including a rotating arch consisting of three FODO cells. Each cell includes four rotating magnets with a maximum magnetic field of 1.8 T and a straight section consisting of one FODO cell. The total length of Nuclotron is 251 m.

The main disadvantage of Nuclotron as an injector into the collider and an independent ring is the shortness of the drift intervals on which the appropriate equipment for these two functions should be installed. All subsequent versions for the Nuclotron upgrades are considered from the point of view of solving this problem.

NUCLOTRON UPGRADES

Two Nuclotron upgrades are considered: Nuclotron 2.1 and Nuclotron 2.2. The former is a magneto-optical structure that preserves all the properties of the existing accelerator and additionally creates conditions for conducting experiments on measuring the EDM of the deuteron by placing electric deflectors or Wien filters on the ring. Each of these devices makes it possible to maintain the direction of the spin vector along the direction of the beam of polarized particles along the accelerator ring.

Solving this problem requires solving four problems within the framework of the eight superperiodic structures $N_{\text{super}} = 8$. We need to implement the concept of a quasi-frozen spin in the proposed optics by increasing the length of the inter-arch straight intervals, as well as to ensure zero dispersion on straight sections and, if possible, to maintain the length of the Nuclotron ring with account for the placement of the required equipment in the tunnel that currently exists.

NICA complex

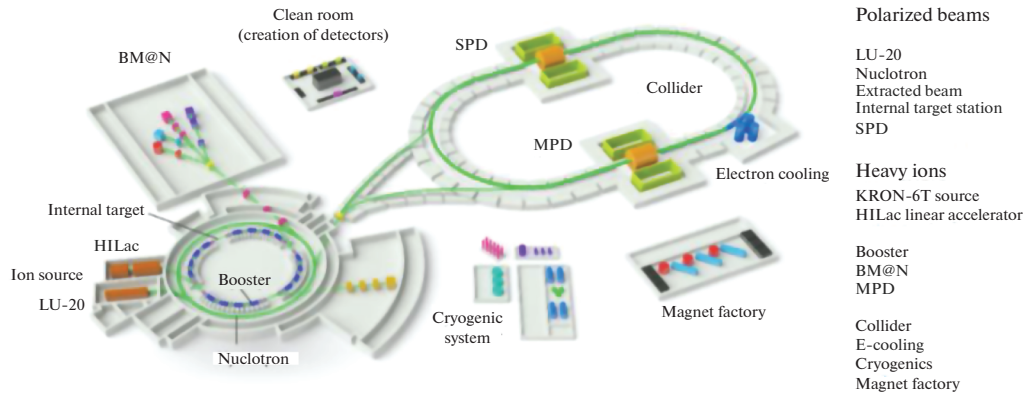


Fig. 2. NICA acceleration complex.

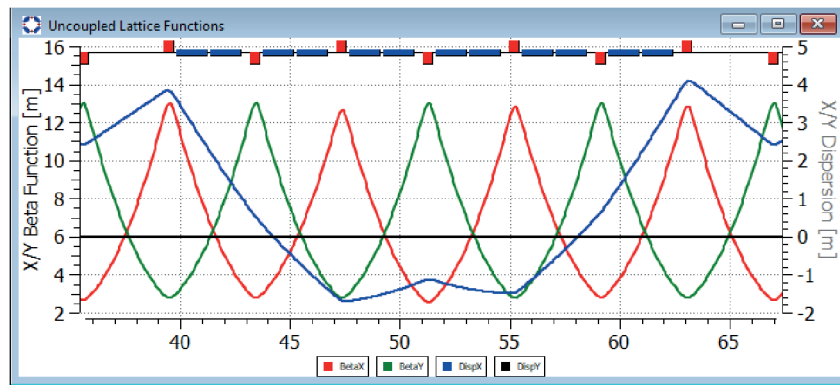


Fig. 3. TWISS functions of one Nuclotron superperiod.

The task of searching for an axion within the framework of this concept can also be included. Currently, there is no complete understanding of how to deal with systematic errors in the experiment proposed. When it comes to Nuclotron 2.1, all the main parameters of Collider, its critical energy, and the electron cooling parameters remain unchanged. This creates serious problems in the relationship between Collider and Nuclotron because the transmission of the cooled beam through the critical energy leads to an irreversible deterioration of the beam parameters, thereby reducing the final luminosity.

In the case of Nuclotron 2.2, it is proposed to make changes in terms of coordination with Collider. The injection energy of polarized protons into Collider remains unchanged at the same level (2–3 GeV). But, in order to exclude the influence of the transmission of critical energy after electron cooling, we could raise the critical energy in Collider above its maximum energy. Other changes include lowering the maximum energy of heavy ions and polarized particles in Nuclotron 2.2. This may expand the functionality of Nuclotron

in its application in experiments on EDM and axion due to the additional space on the ring for placing Wien filters. At the same time, lowering the Nuclotron energy obviously has no effect on the properties of the entire complex because Collider can acquire some of the Nuclotron functions and provide energy for protons and a heavy-ion beam in the entire Nuclotron energy range.

NUCLOTRON 2.1: QUASI-FROZEN SPIN OF A DEUTRON

The task of adapting the Nuclotron structure to the requirements for a structure for measuring the EDM of the deuteron can be defined by a set of problems: to increase straight sections, to suppress dispersion on straight sections, and to maintain the spin direction along the ring.

The first problem, increasing the straight sections to a required length, is solved by enlarging the maximum magnetic field in bending magnets up to 1.8 T and combining two adjacent magnets between quad-

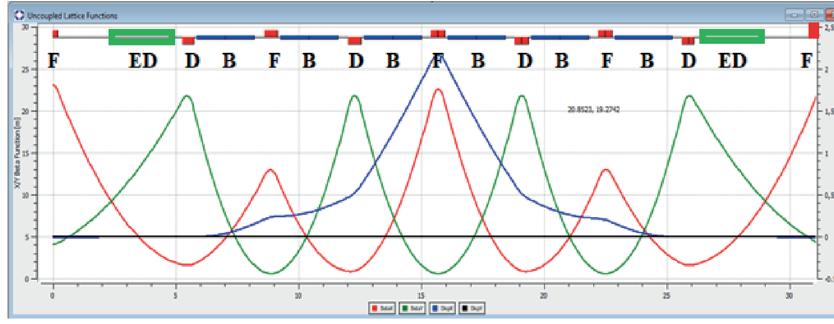


Fig. 4. TWISS functions of one superperiod of upgraded Nuclotron 2.1 with two electrostatic deflectors.

rupole lenses into a single magnet. While solving the first problem, we suppress the dispersion by choosing the phase incursion of radial oscillations on the arches. The proposed upgraded structure of one superperiod of Nuclotron 2.1 is shown in Fig. 4. The main parameters of the upgraded structure of Nuclotron 2.1 are as follows: total length 251 m; betatron frequencies $Q_x = 9.78548$ and $Q_y = 10.6839$; orbital compaction coefficient 0.0134394; slip coefficient -0.0222388 ; natural chromaticity -16.0394 (horizontal) -17.8984 (vertical); defocusing quadrupole (D); focusing quadrupole (F); magnetic dipole magnet (B); electrostatic deflector (ED). All parameters of different Nuclotron versions are summarized in Table 1.

Due to an increase in the magnetic field up to 1.8 T in the bending magnets, the length of a straight gap in each superperiod changes from 7.3 to 10.5 m. The third problem of maintaining the spin direction relative to the momentum required for diagnosing the EDM signal of deuterons is solved by introducing electrostatic deflectors of negative curvature in each superperiod of the structure (Fig. 5).

Within the quasi-frozen spin concept, the magnetic and electric fields are distributed into different elements, which obviously makes the spin direction oscillate relative to the direction of motion, while the spin in the magnetic and electrostatic rotating elements rotates in different directions. We consider the small value of the deuteron magnetic moment anomaly $G = -0.142$ and assume that the spin oscillates relative to the momentum on each magnetic arch of the super-

period within half the spin phase incursion $\Phi_s = +\frac{1}{2N}2\pi\gamma G$, each time returning in the electrostatic deflector by angle $-\Phi_s$, where N is the superperiodicity of the arches. This allows to generally maintain the spin direction along the momentum in the entire ring within the framework of the quasi-frozen spin concept in the accelerator.

In the case of Nuclotron, $N = 8$. Due to the small value of Φ_s , the effective contribution to the expected EDM signal decreases by a factor

$$J_0(\Phi_s) \approx 1 - \frac{(\Phi_s)^2}{4}, \quad (7)$$

which implies a decrease in the EDM of the signal by only several percent.

Let us consider this problem in more detail in order to determine the parameters of electrostatic deflectors in the case of quasi-frozen spin of deuterons. Here, the structure should consist of two different parts (Fig. 5): (1) magnetic arches with bending magnets changing the direction of particle motion on each arch by the angle $\Phi^B = (\pi + 2\alpha)$ and providing spin rotation in the horizontal plane relative to the momentum by the angle $\Phi_s^B = v_s^B \Phi^B$; (2) electrostatic arches with electric deflectors of negative curvature turning the beam on each arch by the angle $\Phi^E = -2\alpha$ and providing spin rotation in the horizontal plane relative to the momen-

Table 1. Nuclotron upgrade parameters

Structure	Magnet length, m	Number of magnets	Total length of magnets, m	$B_{\max}$, Tesla	$B_{\max}R$, T m	Maximum deuteron energy, GeV	Energy per nucleon	Maximum proton energy
Nuclotron	1.44	96	138.24	1.8	39.6	10.14	5.07	10.97
Nuclotron 2.1	2.35	48	112.77	1.8	32.2	8	4	8.79
Nuclotron 2.2.1	1.78	48	85.3	1.8	24.4	5.7	2.85	6.47
Nuclotron 2.2.2	1.78	48	85.44	1.8	24.5	5.7	2.85	6.47
Nuclotron 2.2.3	2.10	48	101	1.8	28.93	7.0	3.5	7.8

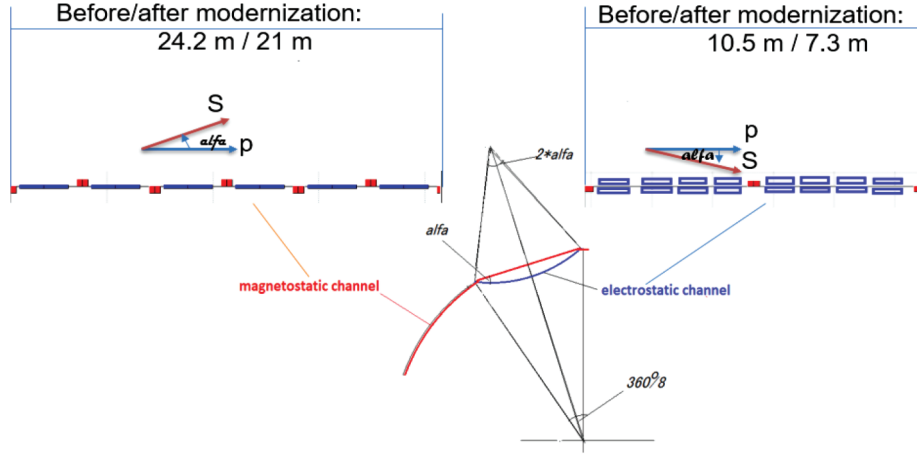


Fig. 5. Electrostatic insert for compensation of spin rotation in magnetostatic arch.

tum in the opposite direction by the angle $\Phi_s^E = v_s^E \Phi^E$. The quasi-frozen spin concept requires that $\Phi_s^E = -\Phi_s^B$. The spin in the electrostatic deflector rotates relative to the momentum with a frequency severalfold greater than in the magnetostatic structure, which is why the following basic relation for two different structural arcs is written:

$$v_s^B \left(\frac{2\pi}{N} + 2\alpha \right) = v_s^E 2\alpha \quad \text{and} \quad (8)$$

$$\alpha = \frac{1}{v_s^E/v_s^B - 1} \frac{\pi}{N} = -\gamma^2 \frac{G}{G+1} \frac{\pi}{N}.$$

If we know the assumed parameters of the bending magnets and the field $E_{ED} = 130$ kV/cm in the electrostatic deflector, then it becomes possible to calculate v_s^E/v_s^B , α , and the curvature radius of the electrostatic deflector $R_{ED} = \frac{\kappa}{E_{ED}}$, where $p_0 = \gamma m \beta c$ and $\kappa = p_0 \beta c / e$ denotes the elastance, which means that $R_{ED} = \frac{mc^2}{e} \frac{\gamma \beta^2}{E_{ED}}$. In this case, the deflector length is determined as follows: $L_{ED} = 2\alpha R_{ED} = -2 \frac{\gamma^2 G}{G+1} \frac{\pi}{N} R_{ED}$. Figure 5 shows how electrostatic deflectors are inserted to implement the quasi-frozen structure of the Nuclotron ring. If the energy $W = 270$ MeV, then $\alpha = 0.026\pi$ and the required length of the electrostatic channel is 7.3 m. In this case, the value of the EDM signal $S_{EDM} = 1 - \alpha^2/4 \approx 0.998$ decreases by 0.2%.

The electrostatic arch of negative curvature assumes the placement of the necessary equipment on a straight section between two magnetic arches, through which the beam is transmitted when the electrostatic deflectors are switched off.

NUCLOTRON 2.2: QUASI-FROZEN SPIN WITH WIEN FILTERS FOR DEUTERONS

In the upgraded version of the magneto-optical structure of Nuclotron 2.2, we introduced a small vertical magnetic field of ~ 100 mT to compensate for the Lorentz force of the electric field. This represents a transition to straight electrostatic deflectors. In the case of adding a magnetic field to the deflector, the newly formed E+B element can be fully classified as a Wien filter. In this case, the ring structure consists of arcs connected by rectilinear sections. In these sections, we placed Wien filters with crossed electric and magnetic fields, ensuring the rectilinear motion of particles due to the zero Lorentz force with simultaneous compensation for the spin rotation that occurred in the magnetic arcs. The total length of all Wien filters is determined by the beam parameters and the requirement to compensate for the spin rotation on the arches, and their number is determined by the convenience of their placement along the ring.

In the magnetic arc with a total number of N magnetic arcs, the particles rotate by the angle $\Phi_{arc}^B = 2\pi/N$, with simultaneous MDM of the spin rotation due to the MDM in the horizontal plane relative to the momentum by the angle $\Phi_s^{arc} = \gamma G \Phi_{arc}^B$. The Wien filters with the E and B fields provide the MDM of the spin rotation in the horizontal plane in the opposite direction relative to the momentum compared to the bending magnets on the arches. In the electrostatic field E_{wf} , the rotation angle is

$$\Phi_{swf}^E = - \left(\gamma G + \frac{\gamma}{\gamma+1} \right) \beta^2 \Phi_{wf}^E, \quad (9)$$

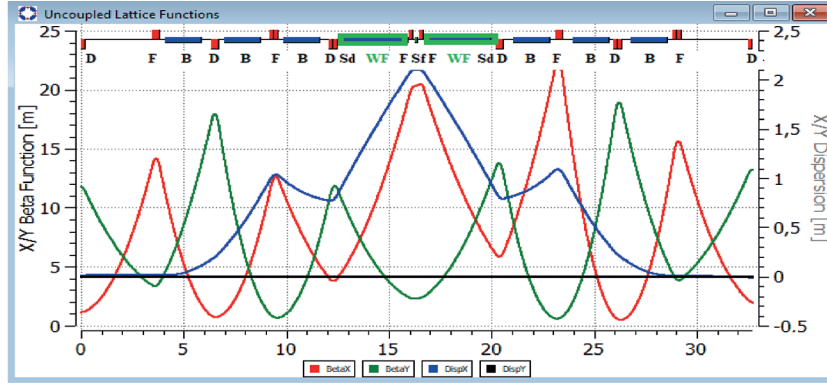


Fig. 6. TWISS functions of one superperiod of upgraded Nuclotron 2.2.1 with two Wien filters.

where $\Phi_{wf}^E = \frac{eE_{wf}}{m_0c^2\gamma\beta^2} L_{wf}$ is the momentum rotation in the electric field of the Wien filter; in the magnetic field B_{wf} , the rotation angle is

$$\Phi_{swf}^B = (\gamma G + 1) \Phi_{wf}^B, \quad (10)$$

where $\Phi_{wf}^B = \frac{eB_{wf}}{m_0c\gamma\beta} L_{wf}$ is the momentum rotation in the magnetic field of the Wien filter of length L_{wf} .

Since the Lorentz force in the Wien filter is zero, the angles $\Phi_{wf}^E = \Phi_{wf}^B$ should be equal to each other. Therefore, they can be defined through one of them. For example, we can use the electric field and the filter length E_{wf} and L_{wf} , which are located on all N straight sections, respectively. As opposed to the frozen spin version, where the total frequency of spin rotation relative to the momentum in the magnetic and electric fields of the rotator with $E + B$ fields is zero, the “quasi-frozen” spin option requires that the total angle of spin rotation per revolution in the E field and in the B field is zero. Thus, the quasi-frozen spin concept requires the fulfillment of the condition $\Phi_{wf}^B - \Phi_{wf}^E = \Phi_{arc}^B$, i.e.,

$$(\gamma G + 1) \Phi_{wf}^B - \left(\gamma G + \frac{\gamma}{\gamma + 1} \right) \beta^2 \Phi_{wf}^E = \gamma G 2\pi / N. \quad (11)$$

These transformations yield a basic relationship for the parameters of one Wien filter of the direct element:

$$L_{wf} E_{wf} = \frac{2\pi}{N} \frac{G}{G + 1} \frac{m_0 c^2}{e} \beta^2 \gamma^3 \quad \text{and} \quad B_{wf} = -\frac{E_{wf}}{c\beta}. \quad (12)$$

Here L_{wf} is the length of one Wien filter. In this case, the total length of all Wien filters $L_{wf}^\Sigma E_{wf} =$

$2\pi \frac{G}{G + 1} \frac{m_0 c^2}{e} \beta^2 \gamma^3$ is a quantity that depends only on the particle type and energy.

Equations (8) and (12), which were obtained first for the electrostatic deflectors and then for the Wien filters, show that the deflecting element type is irrelevant. Thus, with an unchanged magnetic arch, the Wien filter length is smaller by the total kicker length because since it combines the functions of the electrostatic deflector and of the kicker in a single element.

Figure 6 shows the TWISS functions of one superperiod of the first version of the upgraded Nuclotron 2.2.1. Two Wien filters, which restore the deflection of the spin vector direction after the transmission of the magnetic semi-arch, are placed in the central straight sections of the superperiod. The value of the deflection angle of the spin vector from the direction of the pulse $\alpha = \gamma G 2\pi / N$ characterizes the degree of difference between the quasi-frozen magneto-optical structure and the frozen one. In this structure, the number of arches determining the spin vector deflection is equal to the number of superperiods $N = 8$.

If 130 kV/cm is accepted as the maximum electrostatic field, then the magnetic field should be below 80 mT. Such a low magnetic field opens up prospects for simplifying the overall design of the Wien filter. In particular, it is possible to use a permanent magnet or an electric coil with an air core. With such values of the electric and magnetic fields and a deuteron beam energy of 270 MeV, the total length of the Wien filters on all straight sections is about 52 m. The length of a Wien filter is 3.25 m. At an energy of 270 MeV for deuterons (135 MeV per nucleon), the maximum deflection angle of the spin vector is $\alpha \sim \pm 3^\circ$ along the length of a single arc, which characterizes a fairly good approximation to a regular “frozen” structure.

Another upgrade (2.2.2) assumes the spacing of the Wien filters into four straight sections of the superperiod (Fig. 7).

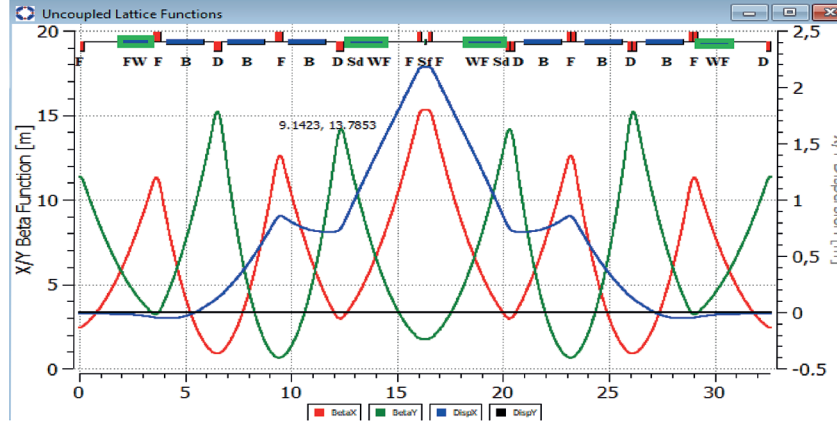


Fig. 7. TWISS functions of one superperiod of upgraded Nucloron 2.2.2 with four Wien filters.

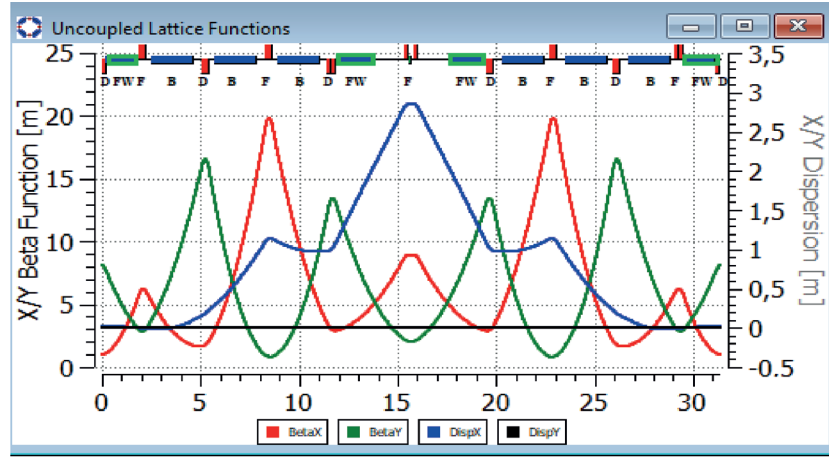


Fig. 8. TWISS functions of one superperiod of upgraded Nucloron 2.2.3 with four Wien filters.

As a result, each superperiod arch is divided into two arches; from the point of view of the bending magnets, the structure becomes $N = 16$ superperiodic while maintaining the overall superperiodicity 8 of the magneto-optical structure with account for all the elements. This solves two problems. Firstly, the ring becomes shaped as a heccaedecagon, so it is rounder and fits better into the existing tunnel. Secondly, with the bending magnet being 1.77-m-long and the Wien filter being 1.65-m-long, we can obtain the deflection angle of the spin vector equal to $\alpha \sim \pm 1.5^\circ$, which is twofold smaller as compared to Nuclotron 2.2.1. This brings the properties of the quasi-frozen and frozen spin structures significantly closer.

To increase the energy, we considered the Nuclotron 2.2.3 upgrade with fewer empty gaps and longer magnets (Fig. 8). In this structure, the maximum energy of deuterons is 3.5 GeV per nucleon and that of protons is 7.8 GeV, which is quite close to the original Nuclotron version.

NUCLOTRON 2.2.2: QUASI-FROZEN SPIN WITH WIEN FILTERS FOR PROTONS

It can be naturally assumed that, with a magnetic superperiodicity of 16, one can use it for low-energy protons, while the main energy limitation here is the free space for Wien filters, which is equal to 52 m here. The maximum possible proton energy is determined by the expression derived from Eq. (12):

$$\gamma(\gamma^2 - 1) = L_{wf} \frac{eE_{wf}}{mc^2} \frac{N}{2\pi} \frac{G_p + 1}{G_p}. \quad (13)$$

Here $L_{wf}N = 52$ m and $G_p = 1.798$ denotes the proton magnetic moment anomaly. Solving the problem relative to γ , we obtain $\gamma = 1.08046724$, which corresponds to the energy $W_p \sim 80$ MeV. At this energy, the spin vector deflection in the magnetic arch is determined by the factor $\gamma G_p = 1.93711246$, and the deflection angle of the spin vector is $\alpha = \pm \frac{1}{2} \frac{\gamma G_p}{16}$, which is

$\alpha = 3.5^\circ$ in numerical expression. Protons can be studied using the same structure as for deuterons, but at a lower energy and with Wien filters turned through 180 degrees, due to the different signs of the anomalous magnetic moments of the particles. In other words, at a proton beam energy of 80 MeV, the behavior of the structure practically coincides with the frozen spin version. Figure 9 shows calculation results for the analyzing power $p\text{-C}^{14}$ in terms of the “average analyzing power” borrowed from [10].

Figure 9 shows that the analyzing power in the vicinity of 100 MeV is close to 0.2, which is 0.35 of the maximum value $A_c = 0.53$ at an energy of 230 MeV. This creates possibilities for studies of the EDM of protons at this energy in the upgraded version of Nuclotron 2.2.2 together with full-scale measurements of deuterons.

The search for galactic axions is possible using the same equipment as that used to search for the EDM. The variation of the spin precession frequency at a constant energy of particles in the ring turns the booster with Wien filters into a broadband antenna used to search for axions by a resonant full or partial spin flip in the oscillating pseudomagnetic field of the axion halo of our galaxy [11]. An increase in horizontal polarization caused by axions is distinguished by the Fourier analysis of the oscillating vertical asymmetry in the internal polarimeter [12].

MAXIMUM ENERGY OF HEAVY PARTICLES AND PROTONS IN THE UPGRADED STRUCTURE OF THE QUASI-FROZEN SPIN OF NUCLOTRON 2.2.2

Calculations have been obtained indicating that experiments on searching for the EDM of deuterons and protons in Nuclotron 2.2.2 require freeing up some space to place Wien filters with a total length of 52 m. Obviously, while maintaining the total length of the tunnel at 251 m, this is possible by either expanding the field in the bending magnets, which we have already assumed to be 1.8 T, or by reducing their total length.

We assumed that the maximum field in the bending magnets cannot exceed 1.8 T and chose the second option, which undoubtedly leads to a decrease in the maximum particle energy in Nuclotron.

MAGNETO-OPTICAL STRUCTURE OF THE UPGRADED NUCLOTRON

The construction of the magneto-optical structure of the upgraded Nuclotron has several objectives. Firstly, the superperiod is constructed in such a way that the central cell has a drift space without bending magnets in the region where the dispersion function has a maximum value at an infinite value of the trajec-

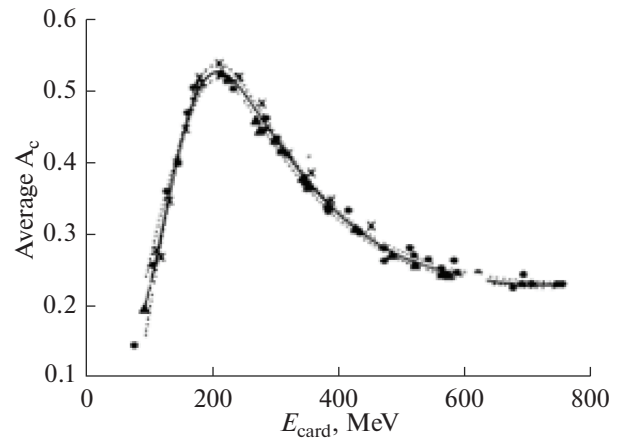


Fig. 9. Plots showing the discrete energy (dots) and the continuous energy (solid line) $\overline{A_c}$ for $\Delta_{lab} = 5^\circ - 20^\circ$. Data points are taken from SIN (Δ), TRIUMF ($\times$), and LAMPF ($\circ$).

tory curvature, thereby allowing to increase the critical energy:

$$\alpha = \frac{1}{C} \int_C \frac{D(s)}{\rho(s)} ds. \quad (14)$$

Secondly, the number of superperiods s , which is equal to eight, slightly exceeds the betatron oscillation frequency ν_x in the horizontal plane, which facilitates the adjustment of the orbital compaction coefficient:

$$\alpha = \frac{\nu^3}{R} \sum_{k=0}^{\infty} \frac{a_k^2}{\nu_x^2 - s^2}. \quad (15)$$

Thirdly, the introduction of the central drift makes it simpler to place the three families of sextupoles needed to regulate the betatron and spin chromaticities, while the lateral drift regions are convenient for placing the Wien filters.

Finally, the maximum dispersion value at the center of the superperiod coincides with the minimum value of the horizontal function β_x .

CONCLUSIONS

When Nuclotron is used as a booster for Collider, the final energy of Nuclotron can be reduced to 2–3 GeV with strongly deteriorating any properties of the NICA complex because all physical experiments with polarized beams are transferred to Collider. This solution has another advantage: reduction of the total length of bending magnets, e.g., by reducing the number of sections in each bending double magnet of Nuclotron to one and by saving space in orbit with a total length of ~ 70 m for additional equipment (particularly, for solenoids that maintain polarization). Moreover, this solution provides obvious opportuni-

ties for additional placement of equipment for searching for the electronic dipole moment and the axion directly in Nuclotron.

The collider experiments impose requirements primarily on luminosity, which impose restrictions, among other things, on the longitudinal phase size of the final plume. During acceleration, it is necessary to overcome the critical energy and divide the beam into 22 plumes using high-frequency field gymnastics without inflating the phase volume.

The possible scheme of the critical energy jump for NICA is considered. The characteristic values are the jump value $\Delta\gamma_{tr} = 0.09$ and the variation rate of the critical energy $d\gamma_{tr}/dt = 8.5 \text{ s}^{-1}$. For the harmonic high-frequency field, the proposed jump has little effect on the longitudinal dynamics due to the smallness of the jump and its low rate as compared to the acceleration rate. For the barrier high-frequency field, the limitation on the jump value yields a threshold value for the number of particles in a uniform plume due to microwave instability and also prevents from reaching the number of particles of 1×10^{12} in the final plume to achieve maximum luminosity.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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